Classification of semisimple Lie algebras

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Abstract

This seminar is meant to be an introduction to the classification of semisimple Lie algebras through root systems and Dynkin diagrams, following sections of Parts II and III of [Hum72]. Since we don't have the time to develop the entire theory we need, some technical results will be stated without proof: this way we will be able to analyse more in detail the theorem giving the isomorphism between two Lie algebras with the same root system.

This seminar is held as final exam for the course "Mathematical Physics — Differential Geometric Methods", with the supervision of Prof. E. Pagani.

1 Introduction

We start by recalling some introductory definitions and properties about Lie algebras we will be needing later.

Definition 1.1 (Lie algebra)**.** A *Lie algebra* over a field $\mathbb{K}$ is a $\mathbb{K}$-vector space L endowed with an operation $[\cdot,\cdot] : L \times L \to L$, called *Lie bracket* or *commutator* such that

(1) the bracket is bilinear;

(2) $[x, x] = 0$ for any $x \in L$

(3) $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ for any $x, y, z \in L$ (*Jacobi identity*).

It is clear that, if $\operatorname{char}\mathbb{K} \neq 2$, properties (1) and (2) imply that the Lie bracket is skew-symmetric. This will always hold as we will be assuming $\mathbb{K}$ to be an algebraically closed field of characteristic 0.

Example 1.2. Given any $\mathbb{K}$-vector space V, the collection of all its endomorphisms $\operatorname{End} V$ is obviously a $\mathbb{K}$-vector space as well. If we define the Lie bracket as the commutator of the composition $[f, g] = f \circ g - g \circ f$, then a simple computation proves that $\mathfrak{gl}(V) := (\operatorname{End} V, +, [\cdot,\cdot])$ is a Lie algebra.

A prototype that will be necessary in our context is the subalgebra of $\mathfrak{sl}(V) \leq_{\textsf{LieAlg}} \mathfrak{gl}(V)$ of endomorphisms with zero trace.

For any $x \in L$, the map obtained by a partial application of the Lie bracket

$$\begin{array}{rccc} \operatorname{ad} x : & L & \longrightarrow & L \\ & y & \longmapsto & [y, x] \end{array}$$

is can be proved by (3) and skew-symmetry of the bracket to be a derivation: this yields that the map $\operatorname{ad} : L \to \operatorname{Der} L$ is defined and is called the *adjoint representation* of L. A Lie algebra

is called *Abelian* if the Lie bracket is always trivial or, equivalently, if the adjoint representation is the zero map.

As for rings, a subspace I of a Lie algebra is called an *ideal* if $[x, y] \in I$ whenever $x \in L$ and $y \in I$ (or the converse, since the bracket is skew).

Given any $X \subseteq L$, the *centraliser* of X in L can be defined as

$$C_L(X) := \{y \in L \mid [y, X] = 0\}.$$

1.1 Semisimplicity

As for groups, we can define the derived series of a Lie algebra L by setting inductively

$$\begin{cases} L^{(0)} = L \\ L^{(k+1)} = [L^{(k)}, L^{(k)}] \quad k \geq 0. \end{cases}$$

Moreover, L is called *solvable* if the derived series "reaches" 0. One can easily prove that every Lie algebra has a unique maximal (with respect to the inclusion) solvable ideal, called the *radical* of L, denoted $\operatorname{Rad} L$.

Definition 1.3 (Simplicity and semisimplicity)**.** A Lie algebra L is called

- *simple*, if it has no ideals other than 0 and itself;
- *semisimple*, if $\operatorname{Rad} L = 0$.

It is clear that simplicity implies semisimplicity.

2 Root Systems and their classification

In this section, we will take a short break from Lie theory and give a quick detour on some structures in a fixed finite-dimensional $\mathbb{R}$-vector space V, endowed with a scalar product $(\cdot, \cdot)$. In particular, for a nonzero $\alpha \in V$, we will be denoting by $H_\alpha := \{\beta \in V \mid (\alpha, \beta) = 0\}$ the hyperplane orthogonal to α and by σ_α the reflection with reflecting hyperplane H_α:

$$\sigma_\alpha(\beta) = \beta - \frac{2(\alpha, \beta)}{(\alpha, \alpha)}\alpha, \qquad \beta \in V.$$

A number that will appear quite often throughout this section is $2\frac{(\beta,\alpha)}{(\alpha,\alpha)}$, hence we will denote it simply by $\langle \beta, \alpha \rangle$: notice that this is linear *only in the first entry.*

2.1 Root Systems

Definition 2.1. A *root system* in V is a subset $\Phi \subset V$ such that

(1) Φ is finite, spans V and does not contain 0;

(2) if $\alpha \in \Phi$, the only multiples of α in Φ are $\pm\alpha$;

(3) if $\alpha \in \Phi$, the reflection σ_α leaves Φ invariant.

(4) if $\alpha, \beta \in \Phi$, then $\langle \beta, \alpha \rangle \in \mathbb{Z}$.

The *rank* of a root system is the dimension of the vector space V and, throughout this seminar, it will be denoted by ℓ.

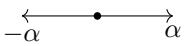
A Root system is called *reducible* if it can be partitioned into the union of two proper subsets such that each root in one subset is orthogonal to each root in the other, and *irreducible* otherwise.

Definition 2.2. The *Weyl group* of a root system Φ is the subgroup of $\mathsf{GL}(V)$ generated by the reflections σ_α, for $\alpha \in \Phi$:

$$\mathscr{W} = \langle \{\sigma_\alpha \mid \alpha \in \Phi\} \rangle.$$

Example 2.3. We show some examples of root systems when $\ell \leq 2$, which can be easily described by drawing a picture.

1. If $\ell = 1$, there is not much choice: the only possibility (modulo isomorphism) is the root system usually labelled A_1, whose Weyl group is $\mathscr{W} = \{\mathrm{Id}, -\mathrm{Id}\} \cong \mathbb{Z}/2\mathbb{Z}$.



2. If $\ell = 2$, there are more possibilities: here are depicted some of them (we will actually see that here we have depicted *all* of them). It is clear that $\mathsf{A}_1 \times \mathsf{A}_1$ is reducible, while the other three are irreducible.

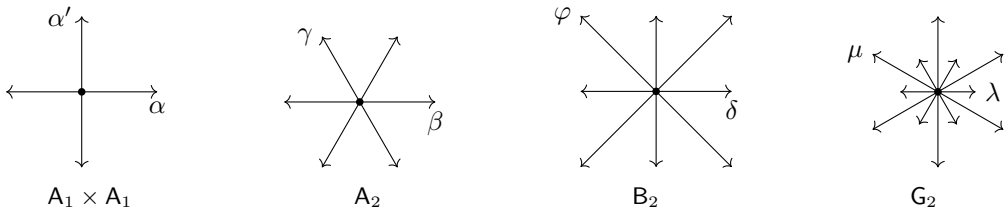


Figure 1: Some root systems in $\mathbb{R}^2$

One quickly notices that condition (4) puts a strong limit on which angles ϑ can there be between any two roots α and β in a root system Φ. By definition of cosine

$$4\cos^2\vartheta = \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \mathbb{Z},$$

hence $\cos\vartheta$ can be either 0 or $\pm\frac{1}{2}$, forcing ϑ to assume the values in the following table.

$\langle \alpha, \beta \rangle$	$\langle \beta, \alpha \rangle$	ϑ
0	0	$\pi/2$
1	1	$\pi/3$
-1	-1	$2\pi/3$
1	2	$\pi/4$
-1	-2	$3\pi/4$
1	3	$\pi/6$
-1	-3	$5\pi/6$

It will be key later that the product $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$ can assume only values 0,1,2 or 3.

Definition 2.4 (Base)**.** A subset $\Delta \subset \Phi$ is called a *base* of the root system Φ if

(1) Δ is a basis of V;

(2) each $\beta \in \Phi$ can be written as a linear combination $\beta = \sum_{\alpha \in \Delta} b_\alpha \alpha$, where the b_α's are all nonnegative or nonpositive.

The roots in a base are called *simple*.

One might ask whether such bases exist: this is indeed not immediate, so we prefer to skip this proof, but it is sufficient to know that there is an explicit way to produce a base for a given root system.

Another crucial property about bases is that the obvious action of the Weyl group on the family of all bases of a fixed root system Φ is transitive: this will be particularly useful in the next section. Moreover, one can also prove that the angle between two simple roots α and β is always obtuse, hence the values $\langle \alpha, \beta \rangle$ and $\langle \beta, \alpha \rangle$ will always be nonpositive. Similarly, it follows that for any two simple roots α and β, then $\alpha - \beta$ cannot be a root.

Each base Δ of a root system Φ comes with an induced ordering on Φ: for any two roots β and γ we write $\beta \prec_\Delta \gamma$ if $\gamma - \beta = \sum_{\alpha \in \Delta} k_\alpha \alpha$ with all non-negative $k'_\alpha s$. In order for this definition to be meaningful, it is crucial for the coefficients k_α in the expression of a root as linear combination of those in a base to be all non-negative or non-positive.

Lemma 2.5. *Let Φ be an irreducible root system and Δ a base. Then there exists a unique maximal root with respect to $\prec_\Delta$.*

2.2 Cartan Matrices and Dynkin diagrams

Consider a basis Δ of a root system Φ and fix an ordering of $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$. Then the matrix $(\langle \alpha_i, \alpha_j \rangle)_{ij}$ is called the Cartan matrix of Φ. At first, this does not seem to be well defined: indeed, thanks to the fact that $\mathscr{W}$ acts transitively on the collection of bases of Φ, one can prove that it depends only on the chosen order of the roots (but this is not too concerning).

For example, here we compute the Cartan matrices of the root systems in Figure 1

$$\mathsf{A}_1 \times \mathsf{A}_1 : \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}; \quad \mathsf{A}_2 : \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}; \quad \mathsf{B}_2 : \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}; \quad \mathsf{G}_2 : \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

We just said that the Cartan matrix is (almost) uniquely determined by the root system: the following proposition states that the converse is true as well, clearly up to isomorphism.

Proposition 2.6. *Let $\Phi \subset V$ and $\Phi' \subset V'$ be two root systems, with bases $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ and $\Delta' = \{\alpha'_a, \dots, \alpha'_\ell\}$ respectively. If $\langle \alpha_i, \alpha_j \rangle = \langle \alpha'_i, \alpha'_j \rangle$ for any $1 \leq i, j \leq \ell$, then there is a unique isomorphism $\varphi : V \to V'$ mapping Φ onto Φ' such that $\varphi(\alpha_i) = \alpha'_i$ and preserving $\langle \cdot, \cdot \rangle$.*

Recall that we've shown how for two distinct roots $\alpha, \beta \in \Phi$, $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \{0, 1, 2, 3\}$. We will be calling the *Coxeter graph* of Φ as the graph with ℓ vertexes, each corresponding to an element of a base of Φ, the i-th and the j-th connected by $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$ edges.

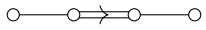
$\mathsf{A}_1 \times \mathsf{A}_1$ $\quad$ A_2 $\quad$ B_2 $\quad$ G_2

In some cases, this is sufficient to retrieve the Cartan matrix of a root system, but if roots have different lengths (such as in B_2 or il G_2), it does not provide the information about which root has which length. To solve this, we add an arrow on multiple edges, pointing towards the edge corresponding to a shorter root.

This improved version of a Coxeter graph is called a *Dynkin diagram*.

$\mathsf{A}_1 \times \mathsf{A}_1$ $\qquad$ A_2 $\qquad$ B_2 $\qquad$ G_2

Example 2.7. Consider the following Dynkin diagram, associated to the root system Φ of type F_4.



then it tells us that Φ has rank 4 and has a basis $\Delta = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ such that

$$\langle \alpha_1, \alpha_2 \rangle \langle \alpha_2, \alpha_1 \rangle = 1; \qquad \langle \alpha_2, \alpha_3 \rangle \langle \alpha_3, \alpha_2 \rangle = 2; \qquad \langle \alpha_3, \alpha_4 \rangle \langle \alpha_4, \alpha_3 \rangle = 1$$

while $\langle \alpha_i, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle = 0$ for all pairs (i, j) not listed above. Moreover, we know that root α_2 is longer than root α_3. Since we know that $\langle \alpha_i, \alpha_j \rangle, \langle \alpha_j, \alpha_i \rangle \leq 0$ whenever i and j are different (this thanks to the angle between two simple roots being always obtuse), then we know all we need to compute the Cartan matrix.

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & \boxed{-2} & 0 \\ 0 & \boxed{-1} & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

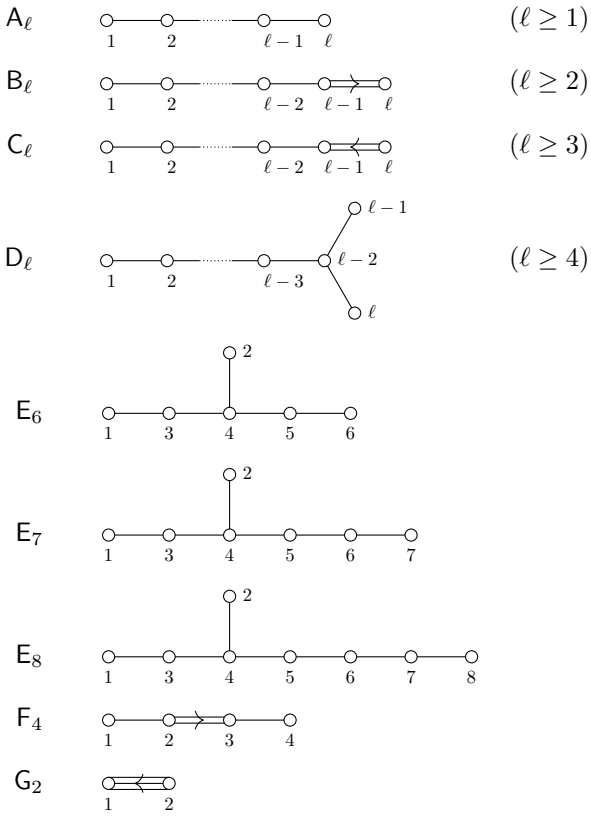
where the -2 is above the diagonal (*i.e.* in column 3) as root α_3 is longer than root α_2. We didn't focus on the diagonal just to have the time to remark that, since it always holds

$$\langle \alpha_i, \alpha_i \rangle = 2 \frac{(\alpha_i, \alpha_i)}{(\alpha_i, \alpha_i)} = 2,$$

then these entries are indeed not so meaningful.

2.3 Classification of irreducible root systems

Theorem 2.8. *If Φ is an irreducible root system of rank ℓ, then its Dynkin diagram is one of the following.*



(2.1)

3 Root Space Decomposition

In order to proceed with root space decomposition, we first need to describe what a toral subalgebra is. To achieve so, we first need to define semisimple elements: this can be done through Jordan decomposition. One can prove that any semisimple Lie algebra can be seen as a subalgebra of $\mathfrak{gl}(V)$ for some vector space V: then an element $x \in L \subseteq \mathfrak{gl}(V)$ is called *semisimple* if the roots of its minimal polynomial are all distinct. If the field is algebraically closed, this turns out to be equivalent to x being diagonalisable. Indeed, by performing Jordan–Chevalley decomposition, any x in L can be written as a sum of two parts $x = x_s + x_n$, where x_s is semisimple and x_n is called the *nilpotent part of* x. One can also prove that $[x_s, x_n] = 0$.

A subalgebra of L consisting only of semisimple elements is called *toral*: such a subalgebra always exists and can be proved to be always Abelian.

Now, a known result in linear algebra (see, for example, [Lan02, Chapter XIV, Exercise 14(d)]) tells us that $\operatorname{ad} H$ being a commuting family of semisimple (diagonalisable) endomorphisms, then there exists a base of common eigenvectors: in particular, L can be decomposed as direct sum of eigenspaces L_α, where

$$L_\alpha = \{x \in L \mid \operatorname{ad} h(x) = \alpha(h)x \ \forall h \in H\},$$

and α ranges over the dual space $H^\vee$ of H. Since L_0 is really just H (we skip the proof as it requires tools we don't have the time to explore) and — L being finite-dimensional — the number of nonzero α's in H is finite, we can write

$$L = C_L(H) \oplus \bigoplus_{\alpha \in \Phi} L_\alpha,$$

where $\Phi = \{\alpha \in H^\vee \mid L_\alpha \neq 0\}$.

The reader can easily see where this is going: we are to show that Φ is indeed a root system prove that it characterises L completely. Before proceeding with this, there is one last piece of machinery we need to introduce.

3.1 The Killing Form

Recall that for an element $x \in L$ we can define its adjoint representation as the map $\operatorname{ad} x \in \operatorname{End} L$ given by $y \mapsto [y, x]$. If we want to study the structure of L, we can look at how its adjoint representation works: one particularly useful tool is given by the Killing form.

Definition 3.1 (Killing form)**.** Let L be any Lie algebra. The Killing form on L is the map $\kappa : L \times L \to \mathbb{K}$ given by $\kappa(x, y) = \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y)$ for $x, y \in L$.

Remark 3.2. It is quite easy to check that the Killing form is a symmetric bilinear form (for symmetry, use the fact that $\operatorname{tr}(f \circ g) = \operatorname{tr}(g \circ f)$ for $f, g \in \operatorname{End}_{\mathbb{K}} V$, *cfr.* [Lan02, Chapter XIII, §1]). It is more difficult to prove that the Killing form is also associative, namely

$$\kappa([x, y], z) = \kappa(x, [y, z]).$$

Theorem 3.3. *Let L be a Lie algebra, then L is semisimple if and only if its Killing form is nondegenerate.*

3.2 Orthogonality, integrality and rationality properties

Remark 3.4. It follows directly from the proof that $L_0 = H$ (which we skipped) that H can be identified with $H^\vee$ via an isomorphism given by the Killing form: to a $\phi \in H^\vee$ corresponds the (unique) element $t_\phi \in H$ satisfying

$$\phi(h) = \kappa(t_\phi, h) \qquad \forall h \in H.$$

This allows us to pull back the killing form to $H^\vee$, defining a new bilinear form

$$\kappa^\vee(\phi, \psi) := \kappa(t_\phi, t_\psi).$$

Furthermore, the restriction of the Killing form to H is nondegenerate.

Proposition 3.5 (Orthogonality)**.**

(1) *Φ spans $H^\vee$;*

(2) *$\kappa(L_\alpha, L_\beta) = 0$ for any $\alpha, \beta \in \Phi$ such that $\alpha + \beta \neq 0$.*

(3) *if $\alpha \in \Phi$, then $-\alpha \in \Phi$;*

(4) *let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Then $[x, y] = \kappa(x, y) t_\alpha$*

(5) *if* $\alpha \in \Phi$, $[L_\alpha, L_{-\alpha}] = \mathbb{K}t_\alpha \leq_{\mathbb{K}} H$ *is one-dimensional;*

(6) $\alpha(t_\alpha) = \kappa(t_\alpha, t_\alpha) \neq 0$, *for* $\alpha \in \Phi$

(7) *If* $\alpha \in \Phi$ *and* x_α *is any nonzero element of* L_α, *then there exists* $y_\alpha \in L_{-\alpha}$ *such that* x_α, y_α *and* $h_\alpha = [x_\alpha, y_\alpha]$ *span a three-dimensional simple subalgebra of* L *isomorphic to* $\mathfrak{sl}(2, \mathbb{K})$ *via*

$$x_\alpha \mapsto \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad y_\alpha \mapsto \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h_\alpha \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix};$$

(8) $h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$; $h_\alpha = -h_{-\alpha}$, $\alpha(h_\alpha) = 2$.

Proof. (1) Assume for a contradiction that Φ fails to span $H^\vee$. Then there exists a nonzero $h \in H$ such that $\alpha(h) = 0$ $\forall \alpha \in \Phi$. This means that $[h, L_\alpha] = 0$ $\forall \alpha \in \Phi$ and, since $[h, H] = 0$, this yields $[h, L] = 0$ or, equivalently, $h \in \mathrm{Z}(L)$. This leads to a contradiction with L being semisimple, though, as by definition of centre $[\mathrm{Z}(L), \mathrm{Z}(L)] = 0$, so the centre is solvable, which cannot be as $\mathrm{Z}(L) \subset \operatorname{Rad} L = 0$.

(2) There certainly exists $h \in H$ such that $(\alpha + \beta)(h) \neq 0$. Then if $x \in L_\alpha$ and $y \in L_\beta$, by associativity of the Killing form

$$\kappa([h, x], y) = -\kappa([x, h], y) = -\kappa(x, [h, y])$$

Which can be expanded as

$$\alpha(h)\kappa(x, y) = \kappa(\alpha(h)x, y) = -\kappa(x, \beta(h)y) = -\beta(h)\kappa(x, y)$$

or, equivalently, $(\alpha + \beta)(h)\kappa(x, y) = 0$, forcing $\kappa(x, y) = 0$.

(3) Let $\alpha \in \Phi$. If $-\alpha \notin \Phi$, and $L_{-\alpha} = 0$ and $\kappa(L_\alpha, L_\beta) = 0$ $\forall \beta \in H^\vee$ (it already vanishes for all $\beta \in H^\vee$ different from $-\alpha$ by (2), and the missing case is here given by $L_{-\alpha} = 0$). This implies that $\kappa(L_\alpha, L) = 0$, a contradiction with the nondegeneracy of κ.

(4) Fix any $h \in H$. Then, by associativity of κ,

$$\kappa(h, [x, y]) = \kappa([h, x], y) = \alpha(h)\kappa(x, y) = \kappa(t_\alpha, h)\kappa(x, y) = \kappa(\kappa(x, y)t_\alpha, h) = \kappa(h, \kappa(x, y)t_\alpha)$$

hence

$$\kappa(H, [x, y] - \kappa(x, y)t_\alpha) = 0.$$

Since the restriction of κ to H is nondegenerate, this implies that $[x, y] - \kappa(x, y)t_\alpha = 0$.

(5) Point (4) shows that either $[L_\alpha, L_{-\alpha}] = 0$ or $[L_\alpha, L_{-\alpha}] = \mathbb{K}t_\alpha$. We show the former cannot hold: let $0 \neq x \in L_\alpha$. If $\kappa(x, L_{-\alpha}) = 0$, then (as explained in the proof of (3)) $\kappa(x, L) = 0$, which cannot hold as κ is nondegenerate, so there must exist some $0 \neq y \in L_{-\alpha}$ such that $\kappa(x, y) = 0$. By (4) this proves $[x, y] \neq 0$.

We skip the proof of points (6), (7) and (8). □

Remark 3.6. It is clear that the h_α defined at point (7) belongs to H. This can be seen, for example, through a quick computation showing that $\operatorname{ad} h \; h_\alpha = 0$ $\forall h \in H$, exploiting the fact that $h_\alpha = [x_\alpha, y_\alpha]$.

We've seen in Remark 3.4 that the Killing form can be transferred to $H^\vee$: up to now, we've worked on an arbitrary algebraically closed field $\mathbb{K}$ of characteristic 0, but in order to work with root systems we need an $\mathbb{R}$-vector space V. To produce this, we observe that, by combining easy linear algebra arguments with Proposition 3.5.(1), every root in Φ can be expressed as a linear combination of roots in a base Δ with rational coefficients. Moreover, if we consider $V_\mathbb{Q} := \operatorname{span}_\mathbb{Q} \Phi \subset H^\vee$, then this is a ℓ-dimensional $\mathbb{Q}$-vector space and the restriction of $\kappa^\vee$ to $V_\mathbb{Q} \times V_\mathbb{Q}$ is a positive-definite symmetric bilinear form. Then our real vector space is obtained by performing extension of scalars $V := V_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{R}$ and equipping it with the obvious extension $(\cdot,\cdot)$ of $\kappa^\vee_{|V_\mathbb{Q} \times V_\mathbb{Q}}$.

Theorem 3.7 (Rationality)**.** *L, H, Φ, V as above. Then*

(1) Φ spans V, and 0 does not belong to Φ;

(2) if $\alpha \in \Phi$, then $-\alpha \in \Phi$ and no other scalar multiple of Φ is a root.

(3) if $\alpha, \beta \in \Phi$, then $\beta - \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha \in \Phi$;

(4) if $\alpha, \beta \in \Phi$, then $\frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z}$.

If we compare this with Definition 2.1, we easily deduce that Φ is indeed a root system in V. This establishes a correspondence between pairs $(L, H) \mapsto (\Phi, V)$, we are now to prove is bijective.

4 Isomorphism Theorem

We will be discussing slightly weaker results concerning simple Lie algebras, as the generalisation to the semisimple case is straightforward.

Proposition 4.1. *Let L be a simple Lie algebra, H a maximal toral subalgebra of L and $\Phi \subset H^\vee$ the set of roots of L relative to H. Then Φ is an irreducible root system in the sense of Definition 2.1.*

Proposition 4.2. *Let L be a semisimple Lie algebra, H a maximal toral subalgebra of L and $\Phi \subset H^\vee$ the set of roots of L relative to H. Fix a base Δ of Φ: then L is generated (as a Lie algebra) by the root spaces L_α, $L_{-\alpha}$ $(\alpha \in \Delta)$ or, equivalently, by root vectors $x_\alpha \in L_\alpha$ and $y_\alpha \in L_{-\alpha}$ $(\alpha \in \Delta)$*

We want to show that, if we consider two pairs (L, H) and (L', H'), each consisting of a Lie algebra and a maximal toral subalgebra, an isomorphism $\alpha \mapsto \alpha'$ of the corresponding root systems produces an isomorphim of the whole algebras. By definition, the said isomorphim comes from an isomorphism $V \to V'$ of the corresponding euclidean spaces, which can be assumed without loss of generality to be an isometry. This extends as well to a unique isomorphism $\psi : H^\vee \to H'^\vee$, which induces an isomorphism $\pi : H \to H'$. We have already produced some crucial constraints we will need to consider in the proof:

- $\pi(t_\alpha) = t'_{\alpha'}$, and therefore $\pi(h_\alpha) = h'_{\alpha'}$
- $\pi(L_\alpha) = L'_{\alpha'}$
- π must preserve the Lie bracket.

In order to proceed freely without taking into account these problems too much, we will be working with a certain choice of simple roots and then extend everything by linearity.

Theorem 4.3. *Let L, L' be simple Lie algebras over $\mathbb{K}$, with respecive maximal toral subalgebras H, H' and corresponding root systems Φ, Φ'. Suppose there is an isomorphism of Φ onto Φ' (denoted $\alpha \mapsto \alpha'$), inducing $\pi : H \to H'$. Fix a base $\Delta \subset \Phi$, so $\Delta' = \{\alpha' \mid \alpha \in \Delta\}$ is a base of Δ'. For each α choose arbitrary algebra isomorphisms $\pi_\alpha : L_\alpha \to L_{\alpha'}$. Then there exists a unique isomorphism $\bar{\pi} : L \to L'$ extending $\pi : H \to H'$ and all the π_α's ($\alpha \in \Delta$).*

Proof.

$\boxed{!}$ If such a map exists, it is clearly unique: each x_α ($\alpha \in \Delta$) determines a unique $y_\alpha \in L_{-\alpha}$ for which $[x_\alpha, y_\alpha] = h_\alpha$, and L is generated by the x_α's and the y_α's by Proposition 4.2.

$\boxed{\exists}$ In order to prove the existence, we exploit the fact that if L and L' are essentially the same, then their direct sum $L \oplus L'$ contains a subalgebra (the "diagonal") isomorphic to L by the projection on either factor. Building such an algebra is quite straightforward: as above, each x_α determines a y_α, so we can use them to define D as the space generated by

$$\tilde{x}_\alpha = (x_\alpha, x'_{\alpha'}), \qquad \tilde{y}_\alpha = (y_\alpha, y'_{\alpha'}), \qquad \tilde{h}_\alpha = (h_\alpha, h'_{\alpha'}) \qquad \forall \alpha \in \Delta.$$

The main problem is therefore to prove D is properly contained in $L \oplus L'$.

Since L and L' are simple, Φ and Φ' are irreducible (Proposition 4.1), thus they have maximal roots β and β' relative to Δ and Δ' resp. (Lemma 2.5), which necessarily correspond under the isomorphism $\Phi \to \Phi'$. Choose now arbitrary nonzero $x \in L_\beta$ and $x' \in L'_{\beta'}$, set $\tilde{x} := (x, x') \in L \oplus L'$, and let M be the subspace of $L \oplus L'$ spanned by all

$$A_{\alpha_1,\dots,\alpha_k}(\tilde{x}) = \operatorname{ad} \tilde{y}_{\alpha_1} \operatorname{ad} \tilde{y}_{\alpha_1} \cdots \operatorname{ad} \tilde{y}_{\alpha_k}\ \tilde{x} \qquad \alpha_i \in \Delta$$

allowing repetitions. First we prove that

$$A_{\alpha_1,\dots,\alpha_k}(\tilde{x}) \in L_{\beta - \sum_i \alpha_i} \oplus L'_{\beta' - \sum_i \alpha'_i} \tag{4.1}$$

by proceeding inductively on the number k of the α_i's. The proof will concern only the first component, but everything can be straightforwardly restated for the second. If $k = 1$, then

$$\begin{aligned}
\operatorname{ad} h \operatorname{ad} y_\alpha\ x &= [[x, y_\alpha], h] = -[h, [x, y_\alpha]] \\
&= [x[y_\alpha, h]] + [y_\alpha, [h, x]] && \text{(Jacobi identity)} \\
&= [x, \operatorname{ad} h\ y_\alpha] - [y_\alpha, \operatorname{ad} h\ x] \\
&= [x, -\alpha(h) y_\alpha] - [y_\alpha, \beta(h) x] && y_\alpha \in L_{-\alpha}, x \in L_\beta \\
&= -\alpha(h)[x, y_\alpha] + \beta(h)[x, y_\alpha] \\
&= (\beta - \alpha)(h) \operatorname{ad} y_\alpha x.
\end{aligned}$$

If k is greater than 1, it is sufficient to repeat the same argument for $\hat{x} = \operatorname{ad} y_{\alpha_2} \cdots \operatorname{ad} y_k\ x \in L_{\hat{\beta}}$ in place of x and $\hat{\beta} = \beta - \sum_{i=2}^k \alpha_i$ in place of β.

In particular, $M \cap (L_\beta \oplus L'_{\beta'})$ is one-dimensional, as any element in $M \cap (L_\beta \oplus L'_{\beta'})$ is a linear combination of terms of the form (z, z') satisfying both

$$\begin{cases} \operatorname{ad} h\ z = \beta(h) z \\ \operatorname{ad} h\ z = (\beta - \sum_i \alpha_i)(h) z; \end{cases}$$

since Δ is a base of H, this implies that the sum must be empty (similarly, this holds as well for the second component) and therefore $M \cap (L_\beta \oplus L'_{\beta'})$ contains only scalar multiples of $\tilde{x}$.

We will now prove that D stabilises M. It is clear by the definition that $\operatorname{ad} \tilde{y}_\alpha(M) \subset M$, so we need to show that the other two families of generators do the same. As usual, we will work componentwise: for the $\tilde{h}_\alpha$'s we proceed inductively. In the base step, a computation similar to the proof of 4.1 shows that

$$\operatorname{ad} h_\alpha \operatorname{ad} y_{\alpha_1} x = [x, \operatorname{ad} h_\alpha \ y_{\alpha_1}] + \operatorname{ad} y_\alpha \operatorname{ad} h_\alpha x = \alpha(h_\alpha) \operatorname{ad} y_\alpha \ x + \beta(h_\alpha) \operatorname{ad} y_\alpha x \in M,$$

h_α being in H and x belonging to L_β. As before, the inductive step is similar, the only difference being that we conclude the second summand belongs to M by use of the induction hypothesis. To show this works for the x_α's, we prove first that $\operatorname{ad} x_\alpha$ and $\operatorname{ad} y_\gamma$ commute whenever $\alpha \neq \gamma$. This follows by the fact that

$$\operatorname{ad} h \ [y_\gamma, x_\alpha] = [[y_\gamma, x_\alpha], h] = [[x_\alpha, h], y_\gamma] + [[h, y_\gamma], x_\alpha] = (\gamma - \alpha)(h)[y_\gamma, x_\alpha] \in L_{\alpha+\gamma}.$$

Since if $\alpha \neq \gamma$, then $\gamma - \alpha$ is not a root (*cfr.* §2.1), so $L_{\gamma-\alpha} = 0$ and

$$\operatorname{ad} x_\alpha \operatorname{ad} y_\gamma \ z = [[z, y_\gamma], x_\alpha] = \cancel{[z, [y_\gamma, x_\alpha]]} + [y_\gamma, [x_\alpha, z]] = [[z, x_\alpha], y_\gamma] = \operatorname{ad} y_\gamma \operatorname{ad} x_\alpha \ z.$$

If $\alpha = \gamma$, an extra summand appears, but this is no big deal:

$$\operatorname{ad} x_\alpha \operatorname{ad} y_\alpha z = [[z, y_\alpha], x_\alpha] + [y_\alpha, [x_\alpha, z]] = \operatorname{ad} h_\alpha z + \operatorname{ad} y_\alpha \operatorname{ad} x_\alpha z.$$

By proceeding inductively, since the base step is given by

$$\operatorname{ad} x_\alpha \ x = [x, x_\alpha] \in L_{\alpha-\beta} = 0,$$

as $\alpha + \beta$ is never a root by maximality of β, we conclude that $\operatorname{ad} \tilde{x}_\alpha$ stabilises M.

Now it is clear that D must be proper: otherwise M would be a proper nonzero ideal of $L \oplus L'$ (as it is stable under the action of $D = L \oplus L'$), but, L and L' being simple, they are the only ideals of this type.

Last, we prove the restrictions of the projections onto the first and second factors of $L \oplus L'$ are isomorphisms. They are obviously homomorphisms, and they are surjective as L (L') is generated by the x_α's and the y_α's (x'_α's and the y'_α's resp., *cfr.* Proposition 4.2), which are contained in the projections of D by the way it is defined. Furthermore, assume D has nonzero intersection with $\ker \operatorname{pr}_{2|D} = L$: then it must contain some element of the form $(w, 0)$ $(0 \neq w \in L)$, so it contains all

$$(\operatorname{ad} z_{\alpha_1} \cdots \operatorname{ad} z_{\alpha_k} \ w, 0), \qquad z_\alpha = x_\alpha \text{ or } y_\alpha, \ \alpha_i \in \Delta$$

which form a nonzero ideal of L, which must coincide with L by simplicity, thus (by symmetry) $D \supset L \oplus L'$: a contradiction. $\lightning$

Finally, we observe that if we define the isomorphism $\bar{\pi} : L \to L'$ as

$$\bar{\pi} = \operatorname{pr}_{1|D}^{-1} \circ \operatorname{pr}_{2|D},$$

then it is clear by construction of D that $\bar{\pi}$ extends π. $\square$

Remark 4.4. Theorem 4.3 establishes a correspondence

$$\left\{\text{Simple Lie algebras}\right\} \rightleftarrows \left\{\text{Irreducible root system}\right\} \rightleftarrows \left\{\text{Dynkin diagrams in (2.1)}\right\}$$

which can be proved extends to a correspondance

$$\left\{\text{Semisimple Lie algebras}\right\} \rightleftarrows \left\{\text{Root system}\right\} \rightleftarrows \left\{\begin{array}{c}\text{Products of}\\ \text{Dynkin diagrams in (2.1)}\end{array}\right\}.$$

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